

Foundations of Policy gradient method

[Policy gradient method]

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi} [\nabla_{\theta} \log \pi_{\theta}(a | s) Q^{\pi}(s, a)]$$

1.0 Content Block

$\gamma V_t + \gamma (R_{t+1} + \gamma V_{t+1} - V_t) \pi_{\theta}(S_{t+1}) - V_t \pi_{\theta}(S_t)$: 2-step TD learning.

2.0 Content Block

$\gamma \sum_{t \leq \tau \leq T} (\gamma \tau - t) (R_{\tau} - b(S_t))$: used by the REINFORCE with baseline algorithm.

3.0 Content Block

Variance reduction REINFORCE is an on-policy algorithm, meaning that the trajectories used for the update must be sampled from the current policy π_{θ} . This can lead to high variance in the updates, as the returns $R(\tau)$ can vary significantly between trajectories. Many variants of REINFORCE have been introduced, under the title of variance reduction.

4.0 Content Block

$\gamma \sum_{n=1}^{\infty} \lambda^{n-1} (1 - \lambda) \cdot (\sum_{k=0}^{n-1} \gamma^k R_{t+k} + \gamma^n V_{t+n} - V_t) \pi_{\theta}(S_{t+n}) - V_t \pi_{\theta}(S_t)$: TD(λ) learning, also known as GAE (generalized advantage estimate). This is obtained by an exponentially decaying sum of the n-step TD learning ones.

5.0 Content Block

Rollout N trajectories in the environment, using π_{θ_t} as the policy function. Compute the policy gradient estimation: $g_i \leftarrow \frac{1}{N} \sum_{n=1}^N \left[\sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta_t}(A_t, S_t, n) \sum_{\tau=t}^T (\gamma \tau R_{\tau} - b(S_t)) \right]$

6.0 Content Block

$\gamma \sum_{t \leq \tau \leq T} (\gamma \tau - t) (R_{\tau} - b(S_t))$: used by the REINFORCE algorithm.

7.0 Content Block

$L(\theta, \theta_i) = \mathbb{E}_{s, a \sim \pi_{\theta_i}} [\pi_{\theta}(a | s) \pi_{\theta_i}(a | s) A_{\pi_{\theta_i}}(s, a)]$ is the surrogate advantage, measuring the performance of π_{θ} relative to the old policy π_{θ_i} .

8.0 Content Block

Policy gradient The REINFORCE algorithm, introduced by Ronald J. Williams in 1992, was the first policy gradient method. It is based on the identity for the policy gradient $\nabla_{\theta} J(\theta) = \mathbb{E} \pi_{\theta} \left[\sum_{t=0}^T \nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) \sum_{\tau=t}^T (\gamma^{\tau-t} R_{\tau}) \mid S_0 = s_0 \right]$ which can be improved via the "causality trick" $\nabla_{\theta} J(\theta) = \mathbb{E} \pi_{\theta} \left[\sum_{t=0}^T \nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) \sum_{\tau=t}^T (\gamma^{\tau-t} R_{\tau}) \mid S_0 = s_0 \right]$ which can be improved via the "causality trick" $\nabla_{\theta} J(\theta) = \mathbb{E} \pi_{\theta} \left[\sum_{t=0}^T \nabla_{\theta} \ln \pi_{\theta}(A_t | S_t) \sum_{\tau=t}^T (\gamma^{\tau-t} R_{\tau}) \mid S_0 = s_0 \right]$

9.0 Content Block

Sutton, Richard S.; Barto, Andrew G. (2018). Reinforcement learning: an introduction. Adaptive computation and machine learning series (2 ed.). Cambridge, Massachusetts: The MIT Press. ISBN 978-0-262-03924-6. Bertsekas, Dimitri P. (2019). Reinforcement learning and optimal control (2 ed.). Belmont, Massachusetts: Athena Scientific. ISBN 978-1-886529-39-7. Grossi, Csaba (2010). Algorithms for Reinforcement Learning. Synthesis Lectures on Artificial Intelligence and Machine Learning (1 ed.). Cham: Springer International Publishing. ISBN 978-3-031-00423-0. Mohamed, Shakir; Rosca, Mihaela; Figurnov, Michael; Mnih, Andriy (2020). "Monte Carlo Gradient Estimation in Machine Learning". Journal of Machine Learning Research. 21 (132): 1–62. arXiv:1906.10652. ISSN 1533-7928.

10.0 Content Block

Formulation Like natural policy gradient, TRPO iteratively updates the policy parameters θ by solving a constrained optimization problem specified coordinate-free: $\{ \max_{\theta} L(\theta, \theta_i) \mid D_{KL}(\pi_{\theta} \parallel \pi_{\theta_i}) \leq \epsilon \}$ where

11.0 Content Block

Use conjugate gradient method to solve for x in $Fx = g$ iteratively without explicit matrix inversion. Use backtracking line search to ensure the trust-region constraint is satisfied. Specifically, it backtracks the step size to ensure the KL constraint and policy improvement. That is, it tests each of the following test-solutions $\theta_{i+1} = \theta_i + \alpha \sqrt{\frac{2\epsilon}{x^T F x}}$, $\theta_{i+1} = \theta_i + \alpha \sqrt{\frac{2\epsilon}{x^T F x}}$, $\theta_{i+1} = \theta_i + \alpha \sqrt{\frac{2\epsilon}{x^T F x}}$, ... until it finds one that both satisfies the KL constraint $D_{KL}(\pi_{\theta_{i+1}} \parallel \pi_{\theta_i}) \leq \epsilon$ and results in a higher $L(\theta_{i+1}, \theta_i) \geq L(\theta_i, \theta_i)$. Here, $\alpha \in (0, 1)$ is the backtracking coefficient.

12.0 Content Block

The natural policy gradient method is a variant of the policy gradient method, proposed by Sham Kakade in 2001. Unlike standard policy gradient methods, which depend on the choice of parameters θ (making updates coordinate-dependent), the natural policy gradient aims to provide a coordinate-free update, which is geometrically "natural".